

配环境

编译器导致的编译失败

1 powershell

先给 **MSVC** 编译器设环境变量(之前以为永远用不着就手动删了)：

<https://superuser.com/questions/1652709/how-to-setup-environment-variable-for-msvc-in-windows-like-gcc>

遇到了 **vscode** 中的 **powershell** 识别不到 **MSVC** 编译器但 **cmd** 能识别的问题：

参阅资料解决：<https://zhuanlan.zhihu.com/p/639866697>

2 cuda

cuda 报错说不支持当前 **MSVC** 的版本，只支持 **2017-2022**，但我装的确实是 **MSVC2022**：

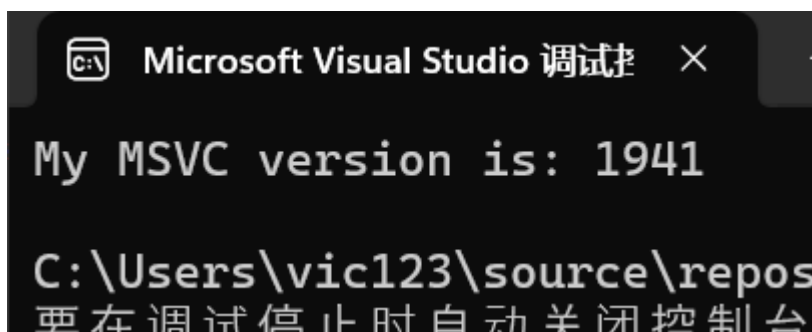
```
ation/build/temp.win-amd64-cpython-38/Release/cuda_rasterizer/backward.obj  
dependency-output D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff-surfel-rasterizer  
:| #error: -- unsupported Microsoft Visual Studio version! Only the versions between 2017 and 2022 (inclusive) are supported
```

查阅网上资料，**MSVC2022** 在设计之初版本号范围是 **v1930-v1939**。因此**cuda** 在检查版本的时候要求版本小于 **1940**，但 **MSVC2022** 的小版本更新太多了，导致 **1940** 也被用作 **MSVC2022** 的版本号：原理：

<https://stackoverflow.com/questions/78515942/cuda-compatibility-with-visual-studio-2022-version-17-10> 解法（未成功，猜测对 **1940** 是有用的，但 **1941** 不行）：<https://github.com/graphdeco-inria/gaussian-splatting/issues/954>

发现我的 **MSVC** 确实属于 **v143**，小版本号是 **1941**，认为和以上 **1940** 情况应该是一致的：

```
Project1 (全局范围)
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num = _MSC_VER; // get the version
6      cout << "My MSVC version is: " << num << endl;
7      return 0;
8  }
```



因此尝试在获取当前 MSVC 版本后通过修改 cuda 的头文件绕开：

```
... | README.md | aaa.txt | host_config.h | fix.md
C: > Program Files > NVIDIA GPU Computing Toolkit > CUDA > v11.8 > include > crt > host_config.h > ...
119
120 #error-- unsupported xlc version! only xlc 13.1 and 16.1 are supported. The nvcc flag '-allow-un
121
122 #endif /* __ibmxl_vrm__ && !(__ibmxl_vrm__ >= 0x0d010000 && __ibmxl_vrm__ < 0x0d020000) && \
123
147 #endif /* __GNUC__ */
148
149 #if defined(_WIN32)
150
151 #if _MSC_VER < 1910 || _MSC_VER >= 1940
152
153 #error -- unsupported Microsoft Visual Studio version! Only the version
154
155 #elif _MSC_VER >= 1910 && _MSC_VER < 1910
156
157 #pragma message("support for this version of Microsoft Visual Studio ha
158
```

3 MSVC

这下 `cuda sdk` 满意了，但 `MSVC` 又不满意了。原来是从 `MSVC 1441` 开始只支持 `12.4` 以上的 `cuda`：

```

C:\Program Files (x86)\Windows Kits\10\include\10.0.26100.0\um" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.26100.0\shared" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.26100.0\winrt" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.26100.0\cppwinrt" "-IC:\Program Files (x86)\Windows Kits\NETFXSDK\4.8\include\um" -c D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff-surfel-rasterization\cuda_rasterizer\adam.cu -o D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff-surfel-rasterization\build\temp.win-amd64-cpython-38\Release\cuda_rasterizer\adam.obj -D__CUDA_NO_HALF_OPERATORS__ -D__CUDA_NO_HALF_CONVERSIONS__ -D__CUDA_NO_BFLOAT16_CONVERSIONS__ -D__CUDA_NO_HALF2_OPERATORS__ --expt-relaxed-constexpr -ID:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff-surfel-rasterization\third_party\glm/ -w -DTORCH_API_INCLUDE_EXTENSION_H -DTORCH_EXTENSION_NAME=_C -D_GLIBCXX_USE_CXX11_ABI=0 -gencode=arch=compute_89,code=compute_89 -gencode=arch=compute_89,code=sm_89
C:/Program Files/Microsoft Visual Studio/2022/Enterprise/VC/Tools/MSVC/14.41.34120/include/yvals_core.h(888): error C: static assertion failed with "error STL1002: Unexpected compiler version, expected CUDA 12.4 or newer."

1 error detected in the compilation of "D:/work/magiclab/gaussian_app/3DGS_SLAM_mobile_app/server/pose2img/submodules/diff-surfel-rasterization/cuda_rasterizer/adam.cu".

adam.cu

[2/7] C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v11.8\bin\nvcc --generate-dependencies-with-compile --dep

```

查阅 `MSVC PR`，提到了可以通过定义一个宏绕开，但还是老老实实选择卸载 `VS 2022` 去装 `VS 2019` 吧

<https://github.com/microsoft/STL/pull/4475>

runtime OS checks when attempting to actually use new machinery.

CUDA

CUDA 12.4.0 is the first version that allows VS 2022 17.10 (`_MSC_VER 1940`), allowing us to remove `--allow-unsupported-compiler` (recently added by [#4392](#)). The last time we increased our requirement was [#2791](#) when CUDA 11.6.0 began supporting VS 2022.

Note that our escape hatch `_ALLOW_COMPILER_AND_STL_VERSION_MISMATCH` remains available if a user can't upgrade their CUDA at the same time as their VS for some reason.

AVX exclusions

装完 `MSVC2019` 后总算不报 `MSVC` 相关的错误了

代码问题导致的编译失败

这个诡异的星号可能用了奇怪的编码格式，导致编译器忽略他和下一行，删了就好饿

```
505     }  
506  
507     // Load Gaussian properties into registers  
508     // ☆  
509     int gaussian_idx = 0;  
510     float2 xy = {0.0f, 0.0f};  
511     float4 nor_o = {0.0f, 0.0f, 0.0f, 0.0f};  
512     float c[C] = {0.0f};  
513     if (valid_splat) {  
514         gaussian_idx = point_list[splat_idx_global];
```

```

D:\work\magiclab\gaussian_app\cuda_extensions\diff_surfel_rasterization\include "-IC:\Program Files (x86)\Microsoft Visual Studio\2019\Enterprise\VC\Tools\MSVC\14.29.30133\ATLMFC\include" "-IC:\Program Files (x86)\Microsoft Visual Studio\2019\Enterprise\VC\Tools\MSVC\14.29.30133\include" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.19041.0\um" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.19041.0\shared" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.19041.0\winrt" "-IC:\Program Files (x86)\Windows Kits\10\include\10.0.19041.0\cppwinrt" -c D:\work\magiclab\gaussian_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu -o D:\work\magiclab\gaussian_app\server\pose2img\submodules\diff_surfel_rasterization\build\temp.win-amd64-cpython-38\gaussian_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.obj -D_CUDA_NO_HALF_OPERATORS__ -D_CUDA_NO_HALF_CONVERSIONS__ -D_CUDA_NO_BFLOAT16_CONVERSIONS__ -D_CUDA_NO_HALF2_OPERATORS__ --xrt-relaxed-constexpr -IC:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\third_party\glm/ -w -DTORCH_API_INCLUDE_EXTENSION_H -D TORCH_EXTENSION_NAME=IBCCXX_USE_CXX11_ABI=0 -gencode=arch=compute_89,code=compute_89 -gencode=arch=compute_89,code=sme_89
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(29): error: expected a declaration
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(29): error: "#" not expected here
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(72): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(73): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(74): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(75): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(76): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(77): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(78): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(79): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(80): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(81): error: unrecognized token
D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff_surfel_rasterization\rasterize_points.cu(82): error: unrecognized token

```

```
8 #define CHECK_INPUT(x) \
9 AT_ASSERTM(x.type().is_cuda(), #x " must be a CUDA tensor")
10 // AT_ASSERTM(x.is_contiguous(), #x " must be contiguous")
11
```

现在是 2024-9-28 1:12 **diff-surfel-rasterization** 终于编译过了

```
(gaussian_app) D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff-surfel-rasterization>pip install .
Looking in indexes: https://mirrors.aliyun.com/pypi/simple/
Processing d:\work\magiclab\gaussian_app\3dgs_slam_mobile_app\server\pose2img\submodules\diff-surfel-rasterization
Preparing metadata (setup.py) ... done
Building wheels for collected packages: diff_surfel_rasterization
  Building wheel for diff_surfel_rasterization (setup.py) ... done
  Created wheel for diff_surfel_rasterization: filename=diff_surfel_rasterization-0.0.1-cp38-cp38-win-amd64.whl size=418628 sha256=382dc2c248db872e68b26f86f72a30f60cc6b37f97e7774f5f9f881117bddd16
  Stored in directory: c:\users\vic123\appdata\local\pip\cache\wheels\d4\20\8c\e5936256507d51e1791c3c97cae384e2d93b18ca20d99f7386
Successfully built diff_surfel_rasterization
Installing collected packages: diff_surfel_rasterization
Successfully installed diff_surfel_rasterization-0.0.1

(gaussian_app) D:\work\magiclab\gaussian_app\3DGS_SLAM_mobile_app\server\pose2img\submodules\diff-surfel-rasterization>
```

跑 **render.py** 需要额外的库

conda install scipy matplotlib open3d lietorch wandb plyfile opencv-python

又寄了 **pytorch3d** 好像只有支持 **py3.9** 的二进制文件 自己编译也没过 这下 **python** 环境要从头来了 早知道当时就重开一个 3.9 不用之前的 3.8 了

重配环境之后还是找不到符合依赖要求的包。一看，草为什么只有给 **linux** 用的的二进制包

检测到当前网页不是中文网页，是否要翻译成中文？

翻译 不翻译 不再提示

pytorch3d / packages / pytorch3d

☆ 5

3d Geometry for pytorch

Conda Files Labels Badges

Filters

Type: All Version: All Label: All

Type	Size	Name	Uploaded	Downloads	Labels
conda	27.5 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt241.tar.bz2	14 days and 3 hours ago	78	main
conda	27.5 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt240.tar.bz2	14 days and 3 hours ago	32	main
conda	27.5 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt231.tar.bz2	14 days and 3 hours ago	25	main
conda	27.4 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt222.tar.bz2	14 days and 3 hours ago	16	main
conda	27.4 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt220.tar.bz2	14 days and 3 hours ago	17	main
conda	27.3 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt212.tar.bz2	14 days and 3 hours ago	17	main
conda	27.3 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt211.tar.bz2	14 days and 3 hours ago	13	main
conda	27.2 MB	linux-64/pytorch3d-0.7.8-py39_cu121_pyt210.tar.bz2	14 days and 3 hours ago	49	main
conda	27.8 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt241.tar.bz2	14 days and 3 hours ago	46	main
conda	27.8 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt240.tar.bz2	14 days and 3 hours ago	16	main
conda	27.9 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt231.tar.bz2	14 days and 3 hours ago	18	main
conda	27.8 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt222.tar.bz2	14 days and 3 hours ago	15	main
conda	27.8 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt220.tar.bz2	14 days and 3 hours ago	20	main
conda	27.6 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt212.tar.bz2	14 days and 3 hours ago	17	main
conda	27.7 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt211.tar.bz2	14 days and 3 hours ago	14	main
conda	27.7 MB	linux-64/pytorch3d-0.7.8-py39_cu118_pyt210.tar.bz2	14 days and 3 hours ago	29	main
conda	27.5 MB	linux-64/pytorch3d-0.7.8-py38_cu121_pyt241.tar.bz2	14 days and 3 hours ago	40	main
conda	27.5 MB	linux-64/pytorch3d-0.7.8-py38_cu121_pyt240.tar.bz2	14 days and 3 hours ago	19	main

<https://anaconda.org/pytorch3d/pytorch3d/files/modal/info/66e452f686b871cc25e673fd>

<https://zhuanlan.zhihu.com/p/609391678> 好好好，自己编译 **pytorch3d** 过了

```
Requirement already satisfied: colorama in d:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages (from tqdm->iopath->pytorch3d==0.7.8) (0.4.6)
Installing collected packages: pytorch3d
  DEPRECATION: Legacy editable install of pytorch3d==0.7.8 from file:///D:/work/magiclab/gaussian_app/3DGS/server/pose2img/pytorch3d (setup.py develop) is deprecated. pip 25.0 will enforce this behaviour change. A possible replacement is to add a pyproject.toml or enable --use-pep517, and use setuptools >= 64. If the resulting installation is not behaving as expected, try using --config-settings editable_mode=compat. Please consult the setuptools documentation for more information. Discussion can be found at https://github.com/pypa/pip/issues/11457
  Running setup.py develop for pytorch3d
Successfully installed pytorch3d
```

又急，lietorch 和当前环境的不知道什么东西不兼容

```
Using cached lietorch-0.6.2.post1-py3-none-any.whl
Installing collected packages: lietorch
Successfully installed lietorch-0.6.2.post1

(gaussian_render_39) D:\work\magiclab\gaussian_app\3DGS\server\pose2img>python render.py
Traceback (most recent call last):
  File "D:\work\magiclab\gaussian_app\3DGS\server\pose2img\render.py", line 4, in <module>
    from gaussian.gaussian_model import GaussianModel
  File "D:\work\magiclab\gaussian_app\3DGS\server\pose2img\gaussian\gaussian_model.py", line 7, in <module>
    from gaussian.gaussian_base import GaussianBase
  File "D:\work\magiclab\gaussian_app\3DGS\server\pose2img\gaussian\gaussian_base.py", line 14, in <module>
    from gaussian.vis_utils import vis_rgbdnua, load_ply, calc_psnr
  File "D:\work\magiclab\gaussian_app\3DGS\server\pose2img\gaussian\vis_utils.py", line 9, in <module>
    from lietorch import SE3
  File "D:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages\lietorch\__init__.py", line 75, in <module>
    _init_backend()
  File "D:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages\lietorch\__init__.py", line 55, in _init_backend
    torch.ops.load_library(libpath)
  File "D:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages\torch\_ops.py", line 1295, in load_library
    ctypes.CDLL(path)
```

摸不着头脑，conda 源里写的 python>=3.10，那我是怎么把 lietorch 安装下来的
No Description

Uploaded	Thu Aug 24 18:17:45 2023
md5 checksum	ff692762e0e0023b1c11e8f7332270ce
build	py_1
build_number	1
depends	numpy, python >=3.10, pytorch, torchvision
license	MIT
noarch	python
subdir	noarch
target-triplet	None-any-None
timestamp	1692900691687

算了也手动编译好了。。。怎么自己编译出 lietorch 版本是 0.2

```
__pycache__.lietorch_backends.cpython-39: module references __file__
__pycache__.lietorch_extras.cpython-39: module references __file__
creating dist
creating 'dist\lietorch-0.2-py3.9-win-amd64.egg' and adding 'build\bdist.win-amd64\egg' to it
removing 'build\bdist.win-amd64\egg' (and everything under it)
Processing lietorch-0.2-py3.9-win-amd64.egg
creating d:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages\lietorch-0.2-py3.9-win-amd64.egg
Extracting lietorch-0.2-py3.9-win-amd64.egg to d:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages
Adding lietorch 0.2 to easy-install.pth file

Installed d:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages\lietorch-0.2-py3.9-win-amd64.egg
Processing dependencies for lietorch==0.2
Finished processing dependencies for lietorch==0.2 ?
```

原来作者是手动管理版本的。。。上次修改版本号还是 2021 年。谴责一下

```
zachteed, 3 years ago | 1 author (zachteed)
from setuptools import setup
from torch.utils.cpp_extension import BuildExtension, CUDAExtension

import os.path as osp

ROOT = osp.dirname(osp.abspath(__file__))

setup(
    name='lietorch',
    version='0.2',
    description='Lie Groups for PyTorch',
    author='teedrz',
    packages=['lietorch'],
    ext_modules=[
        CUDAExtension('lietorch_backends',
            ['lietorch_backends.cpp',
            ])
```


winwin😊

能跑了

```
(gaussian_render_39) D:\work\magiclab\gaussian_app\3DGS\server\pose2img>python render.py
wandb: Using wandb-core as the SDK backend. Please refer to https://wandb.me/wandb-core for more information.
wandb: Currently logged in as: 1582294705 (test-wuke). Use `wandb login --relogin` to force relogin
wandb: WARNING If you're specifying your api key in code, ensure this code is not shared publicly.
wandb: WARNING Consider setting the WANDB_API_KEY environment variable, or running `wandb login` from the command line.
wandb: Appending key for api.wandb.ai to the config file: s\vic123\netrc
wandb: Tracking run with wandb version: Focus folder in explorer (ctrl + click)
wandb: Run data is saved locally in: D:\work\magiclab\gaussian_app\3DGS\server\pose2img\wandb\run-20240928_131348-ftwu1jmu
wandb: Run `wandb offline` to turn off syncing.
wandb: Syncing run smart-oath-142
wandb: View project at https://wandb.ai/test-wuke/Droid2DAcc
wandb: View run at https://wandb.ai/test-wuke/Droid2DAcc/runs/ftwu1jmu
D:\applications\miniconda3\envs\gaussian_render_39\lib\site-packages\torch\functional.py:513: UserWarning: torch.meshgrid: in an upcoming release, it will be required to pass the indexing argument. (Triggered internally at C:\cb\pytorch_100000000000\work\aten\src\ATen\native\TensorShape.cpp:3610.)
  return _VF.meshgrid(tensors, **kwargs) # type: ignore[attr-defined]
wandb: View run smart-oath-142 at: https://wandb.ai/test-wuke/Droid2DAcc/runs/ftwu1jmu
wandb: Find logs at: wandb\run-20240928_131348-ftwu1jmu\logs
```

output >  render.png



资料

解释了编译 **cuda** 为什么要用 **MSVC** <https://stackoverflow.com/questions/57450317/confusion-about-cuda-installation-instructions-what-exactly-is-microsoft-visual>

提到了一种不需要装完整 **visual studio** 编译 **cuda** 的方法 <https://snackshark.blogspot.com/2021/05/setup-cuda-without-visual-studio-oneof.html>